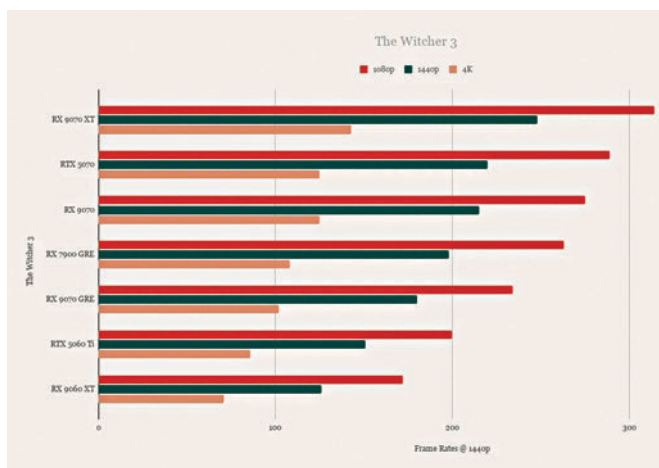


API Performance

Basemark GPU is a nice benchmark to compare the performance of different graphics APIs between cards. We can use the same textures with OpenGL, Vulkan and DirectX 12 to see if the graphics card excels at any particular API more than the rest or if the performance is consistent across the board.



Procyon AI Text Generation

The UL Procyon AI Text Generation benchmark automates local LLM testing by executing standard models, such as Phi-3.5-mini, across diverse real-world prompts. It simulates creative writing and Retrieval-Augmented Generation where the model summarizes text by drawing from external data. The benchmark interacts directly with hardware-specific runtime engines like Microsoft DirectML or NVIDIA TensorRT to isolate real-world optimization. It breaks generation speed into Tokens per Second, initial processing latency, overall resource consumption.

POWER AND THERMALS

Power behaviour is one area where the RX 9070 GRE needs some context. AMD's reference specification lists a 220W total board power, but the test card peaked at 270W during sustained load. That is a meaningful increase over the official figure, although not unusual for partner cards with more aggressive boost behaviour and factory tuning.

Thermals were well controlled, with the GPU peaking at 77 degrees Celsius. That puts the card in a comfortable operating range for a modern gaming GPU and suggests the cooler has enough headroom even when the board is drawing more than the reference power target. The card is not especially frugal when pushed, but it does not appear thermally stressed either. For most users, the more important

takeaway is that case airflow will matter, particularly in compact builds or cabinets with restricted front intake.

OVERCLOCKING

The Radeon RX 9070 GRE leaves some room for tuning, but expectations need to be realistic. This is already a high-clocked RDNA 4 GPU, with an official boost clock of up to 2,920 MHz, and partner cards are likely to arrive with their own factory overlocks. That means manual overclocking is less about transforming the card and more about finding a stable balance between frequency, power, fan noise, and memory behaviour. In practice, the more useful approach is likely to be a mild GPU clock uplift paired with careful power-limit tuning. Memory overclocking may also help in bandwidth-sensitive games, since the card uses a 192-bit bus with 18 Gbps GDDR6. However, pushing too hard can quickly eat into efficiency, especially when the test card is already able to touch 270 W under peak load. For most users, AMD's tuning tools should be treated as a way to refine the card rather than chase large gains. A mild undervolt may actually be the more interesting route, especially if the goal is to reduce fan noise while retaining most of the card's 1440p performance.

VERDICT

The GIGABYTE AMD Radeon RX 9070 GRE is a well-received addition to AMD's

RDNA 4 line-up. It does not replace the RX 9070 XT (or the RX 9070) for users who want the highest frame rates, and it does not make the RX 9060 XT irrelevant for mainstream buyers. Instead, it fills the gap between them with a card that is clearly aimed at 1440p gaming, modern display support, better ray tracing than previous Radeon mid-range cards, and a stronger AI and media feature set than RDNA 3. This is a year-old

card, and we don't know why AMD chose to release it globally now. Perhaps they've got surplus or perhaps it's a smarter stop gap until the next generation.

Compared with the RX 7900 GRE, the AMD Radeon RX 9070 GRE benefits from RDNA 4's newer ray accelerators, second-generation AI accelerators, improved media engine, and newer display connectivity. When compared with the RTX 5060 Ti, it should generally feel like the more capable 1440p gaming card. The main caveat here is power. A 270 W peak draw on the test card is not alarming, but it does make the RX 9070 GRE less efficient than the 220 W reference figure might suggest. Even so, the RX 9070 GRE lands as a practical choice for gamers who want more than entry-level performance without paying full upper-mid-range money. As for Indian pricing, you'll have to wait a day.

—Mithun Mohandas

SPECIFICATIONS

Shader: 3 | Dual Compute Units: 24 | Compute Units: 48 | Shaders: 3072 | Tensor Cores: 96 | AI TOPS: 549 | RT Cores: 48 | Texture Units: 192 | ROP Units: 96 | Base Clock: 1420 | Boost Clock: 2790 | Memory Clock: 2250 | Total Video Memory: 12 GB GDDR6 | TGP: 220W

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